Finite-horizon windowed distributional chaos on the Hilbert cube: a residual theorem and its sharp finite-certificate obstruction

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Abstract

On the Hilbert cube $Q_\infty = [0,1]^{\mathbb{N}}$ the companion paper [1] establishes a residual depth hierarchy of margin-stable covering schemes in the full non-autonomous mapping space $\mathcal{F}(Q_\infty) = C(Q_\infty, Q_\infty)^{\mathbb{N}}$, together with an empty-interior obstruction showing that no open class can certify exact covering in arbitrarily deep coordinates; it leaves open whether the infinite-horizon dynamical conclusions of the underlying dense theorem of [2] — all-tail semifactors, windowed distributional chaos of type 1, infinite entropy and infinite metric mean dimension — are residual. This paper advances that question on its *finite-horizon* layer. First we prove a positive residual theorem: for any countable family of dense-open window sequences there is a comeager set $\mathcal{G} \subseteq \mathcal{F}(Q_\infty)$ such that every system in $\mathcal{G}$ admits finite-horizon windowed DC1 witnesses of every order, at every tail, every rational proximal scale, every rational tolerance, and the fixed separation scale — pairs of orbits proximal on an arbitrarily large prescribed time-fraction of arbitrarily long horizons and separated on a positive time-fraction, both visiting the prescribed windows up to those horizons. The same residual set in fact contains finite scrambled packets of every prescribed cardinality: for every M there are arbitrarily long horizons carrying M orbits which are all mutually proximal on a common large set of times and whose every pair is t_0-separated on a positive fraction of times. The mechanism is a finite-horizon stability lemma whose orbit-tracking modulus does *not* decay with coordinate depth, in contrast with the depth obstruction of [1], combined with the dense windowed-horseshoe theorem of [2], a block schedule, and a binary coding argument across the separation blocks. Second we prove that this is sharp at the level of certificates: every finite-horizon certified open set — one whose membership depends only on a bounded window of coordinate maps — that is contained in the class of systems with an infinite-metric-mean-dimension tail, or in the corresponding scale-law entropy class from [2], is *empty*. The proof is an explicit coordinate squeeze applied past the certifying window, which stays uniformly close to every system in the set yet caps the metric mean dimension of every later tail. The same squeeze also defeats the all-tail symbolic semifactor, while leaving a bare uncountable DC1 set untouched, so the certificate obstruction separates the growth-type conclusions from the cardinality conclusion. We record the Banach–Mazur corollary and an honest ledger: this rules out the natural finite-certificate method class for the infinite-horizon problem, but does not decide whether the full conclusions are comeager; that remains open for methods that are not finite-horizon.

1 Introduction

A non-autonomous discrete dynamical system on a compact metric space (X, ρ) is a sequence $f = (f_1, f_2, \dots)$ of continuous self-maps acting by composition $F^j = f_j \circ \cdots \circ f_1$. The full mapping space

$$\mathcal{F}(X) = C(X, X)^{\mathbb{N}}, \qquad D_X(f, g) = \sup_{n \geq 1} \sup_{x \in X} \rho\big(f_n(x), g_n(x)\big),$$

is complete and hence Baire. On finite-dimensional cubes and on compact manifolds the construction of [2] produces an open dense class of systems with all-tail full complexity: after deleting any finite initial segment one still has a compact symbolic semifactor onto a full variable-alphabet shift, a continuum-sized scrambled set for distributional chaos of type 1 (DC1) visiting prescribed dense open windows at prescribed times, infinite topological entropy, and the sharp small-scale entropy spectrum. On the Hilbert cube

$$Q_\infty = [0, 1]^{\mathbb{N}}, \qquad \rho(x, y) = \sum_{j \geq 1} 2^{-j} \, |x_j - y_j|,$$

the same conclusions hold only *densely*, because coordinate j carries metric weight 2^{-j} and exact control of arbitrarily deep coordinates is not stable. The companion paper [1] analyses exactly this gap. It proves that for every depth N the class of systems carrying a margin-stable depth-N covering scheme is open dense, with an explicit stability radius of order 2^{-N}, so that the intersection over all depths is residual; that on this residual set one-step covering is exact at every finite depth; and, on the negative side, that the dense exact-horseshoe class — indeed any class forcing some image to have full range in infinitely many coordinates — has empty interior. What [1] explicitly leaves open (its Problem, restated below as Problem 6.1) is whether the *infinite-horizon dynamical* conclusions are themselves residual:

> "What is missing is exactly a mechanism producing, for a residual set of systems, orbits threading infinitely many prescribed bricks — equivalently, a replacement for the backward induction [2, Lemma 13.1] that consumes only finite-depth information per step."

This open question traces back to Section 14 of [2], where the "dense, not open" phenomenon on Q_∞ is first recorded and a residual theorem "by a Baire-fusion mechanism" is requested.

This paper

We attack the question on the layer where it is tractable: *finite horizons*. The infinite-horizon DC1 conclusion asks for a scrambled pair (x, y) whose orbits are proximal along a set of times of upper density 1 and separated along a set of upper density 1, with prescribed window visits at *all* times. We weaken "all times" to "all times below a horizon N" and "upper density" to "a prescribed fraction of $\{0, \dots, N-1\}$", and prove that the resulting finite-horizon witnesses exist residually, of every order and at every scale, and then prove that no finite-horizon *certificate* can reach the infinite-horizon conclusions. The two results sit on opposite sides of a sharp line.

Theorem 1.1 (Residual finite-horizon windowed DC1 witnesses; Theorem 4.5 below)**.** *Let $\{W_\bullet^a\}_{a \geq 1}$ be any countable family of sequences of dense open subsets of Q_∞, and fix a separation scale $t_0 \in (0, 1)$. There is a comeager set $\mathcal{G} \subseteq \mathcal{F}(Q_\infty)$ such that every $g \in \mathcal{G}$ has the*

following property: for every index a, every tail $r \geq 0$, every rational proximal scale $\varepsilon \in (0,1)$, and all rational $0 < \theta < \eta < 1$, for every $N_0 \geq 1$ there exists $N \geq N_0$ for which there are points $x, y \in Q_\infty$ and disjoint sets of times $P, S \subseteq \{0, \dots, N-1\}$ with $\#P \geq (1-\eta)N$ and $\#S \geq \theta N$ for which

$$\rho\big(G_r^j(x), G_r^j(y)\big) < \varepsilon \quad (j \in P), \qquad \rho\big(G_r^j(x), G_r^j(y)\big) > t_0 \quad (j \in S),$$

and $G_r^j(x), G_r^j(y) \in W_{r+j+1}^a$ for every $j < N$, where $G_r^j = g_{r+j} \circ \cdots \circ g_{r+1}$ and $G_r^0 = \mathrm{id}$.

The witness in Theorem 1.1 is the finite-horizon, fixed-tolerance shadow of a DC1 scrambled pair: proximal on a $(1-\eta)$-fraction of the horizon, separated at a fixed scale on a positive θ-fraction, inside the prescribed windows up to the horizon. Letting $\eta \downarrow 0$ and then choosing any smaller $\theta > 0$ produces arbitrarily long horizons on which the pair is proximal on an arbitrarily large fraction of times and still separated at scale t_0 on a positive fraction — the finite-horizon content of "proximal and separated of every order."

The same construction also gives finite scrambled packets, not only pairs.

Theorem 1.2 (Residual finite-horizon scrambled packets)**.** *Under the hypotheses of Theorem 1.1, possibly after replacing the comeager set by a smaller comeager set, every g has the following stronger property. For every window sequence $W_\bullet^a$, every tail $r \geq 0$, every integer $M \geq 2$, every rational proximal scale $\varepsilon \in (0,1)$, every rational $0 < \eta < 1$, and every N_0, there are $N \geq N_0$, points*

$$x^1, \dots, x^M \in Q_\infty,$$

a common set $P \subseteq \{0, \dots, N-1\}$, and pair-dependent sets $S_{uv} \subseteq \{0, \dots, N-1\}$, $1 \leq u < v \leq M$, such that

$$\#P \geq (1-\eta)N, \qquad \#S_{uv} \geq \frac{\eta}{8}N \quad (1 \leq u < v \leq M),$$

all M orbit segments visit the prescribed windows up to time N, all pairs are ε-proximal at every time in P, and the pair (x^u, x^v) is t_0-separated at every time in S_{uv}.

Theorem 1.3 (No finite-horizon certificate forces infinite-horizon complexity; Theorem 5.4 below)**.** *Call an open set $V \subseteq \mathcal{F}(Q_\infty)$* finite-horizon certified *if there are an offset $r \geq 0$ and a horizon $N \geq 1$ such that membership $g \in V$ depends only on the finite window $(g_{r+1}, \dots, g_{r+N})$. If V is finite-horizon certified and*

$$V \subseteq \big\{\, g \in \mathcal{F}(Q_\infty) : \ \mathrm{mdim_M}(\mathrm{tail}_s g) = \infty \text{ for some } s \geq 0 \,\big\},$$

then $V = \varnothing$. The same holds if the entropy hypothesis is replaced by the scale-law entropy class of [2]: for some tail and every q, for some constant C_q, the small-scale entropy satisfies

$$h_{\mathrm{top}}^{(\varepsilon)}(\mathrm{tail}_s g) \geq q \log(1/\varepsilon) - C_q$$

for all sufficiently small ε.

Theorem 1.3 says the line drawn by Theorem 1.1 cannot be crossed by the obvious method. A natural strategy for the infinite-horizon problem is to exhibit a countable family of open dense sets, each certified by a bounded window, whose intersection forces infinite-horizon complexity, and to invoke Baire. Theorem 1.3 shows that every individual set in such a family that already entails an infinite-$\mathrm{mdim_M}$ tail, or the scale-law entropy tail supplied by the dense theorem, is empty, hence not dense: the entailment and finite-horizon certification are incompatible. The infinite-horizon conclusions are genuinely non-finite-horizon properties, and any residual proof must read unboundedly many coordinate maps to certify a single system.

Why depth and horizon behave oppositely

The contrast between the two theorems is the engine of the paper, and it is worth isolating. The obstruction of [1] is about *coordinate depth*: a condition that constrains image coordinate ℓ has D-stability only $O(2^{-\ell})$, so margins erode geometrically in depth and no open set survives unbounded depth. By contrast, the finite-horizon witnesses of Theorem 1.1 are stable because their defining inequalities involve the metric ρ *directly*, not individual deep coordinates, and the orbit-tracking modulus over a *fixed* horizon N is finite and depth-free: if $D(h, g) < \delta$ then the two N-orbits stay within $e_N \le \delta + \omega(e_{N-1})$, a quantity that depends on the uniform-continuity moduli ω of the maps and on N, but not on how deep the proximity or separation is read (Lemma 3.2). This is the precise sense in which the finite-horizon layer escapes the depth obstruction: horizons are stable while depths are not. The two obstructions are independent, and the present Theorem 1.3 is a genuinely *horizon* obstruction — the squeeze acts past a fixed window in time, not past a fixed coordinate — complementary to the depth obstruction of [1].

An honest ledger

We are careful about what is and is not proved. Theorem 1.1 produces witnesses that are finite-horizon and carry a *fixed* separation tolerance t_0; it does not produce a single scrambled pair working at all horizons simultaneously, nor does it recover the upper-density-one statements that define DC1 in the limit. Theorem 1.3 rules out the natural finite-certificate route to the infinite-horizon problem, but it does *not* prove that the infinite-horizon class is non-comeager: a residual set need not be a countable intersection of finite-horizon certified open sets. Thus Problem 6.1 (residuality of the full infinite-horizon conclusions) remains open; what we add is a residual finite-horizon theorem on one side and, on the other, a proof that the most direct method class cannot settle it. Section 6 states this precisely.

Organization

Section 2 fixes notation, recalls the Baire property and the inputs we import from [2, 1]. Section 3 defines the finite-horizon windowed DC1 witness and proves its openness and depth-free stability. It also defines finite scrambled packets and proves their openness. Section 4 proves density via the dense windowed-horseshoe theorem, a block schedule, and a binary coding argument across separation times, then assembles Theorems 1.1 and 1.2 by Baire intersection. Section 5 proves the finite-horizon certificate obstruction (Theorem 1.3) and its Banach–Mazur corollary. Section 6 records the ledger and the open problem.

2 Preliminaries

Throughout, $Q_\infty = [0,1]^{\mathbb{N}}$ carries $\rho(x, y) = \sum_{j\ge1} 2^{-j}|x_j - y_j|$, which induces the product topology and makes Q_∞ compact, with $\operatorname{diam}_\rho(Q_\infty) = 1$. We write $\pi_\ell : Q_\infty \to [0,1]$ for the ℓ-th coordinate projection. The space $C(Q_\infty, Q_\infty)$ carries the uniform metric $d_\infty(u, v) = \sup_x \rho(u(x), v(x))$, and

$$\mathcal{F}(Q_\infty) = C(Q_\infty, Q_\infty)^{\mathbb{N}}, \qquad D(f, g) = \sup_{n\ge1} d_\infty(f_n, g_n).$$

For $g \in \mathcal{F}(Q_\infty)$, $r \ge 0$ and $j \ge 0$ put $G_r^0 = \mathrm{id}$ and $G_r^j = g_{r+j} \circ \cdots \circ g_{r+1}$; the *tail at* r is $\mathrm{tail}_r g = (g_{r+1}, g_{r+2}, \dots)$.

Lemma 2.1 ([2, Lemma 2.1])**.** *$(\mathcal{F}(Q_\infty), D)$ is a complete metric space, and hence a Baire space.*

Proof. $C(Q_\infty, Q_\infty)$ is complete because Q_∞ is compact and uniform limits of continuous maps are continuous. If $(f^{(k)})_k$ is D-Cauchy, each coordinate sequence $(f_n^{(k)})_k$ is d_∞-Cauchy with a modulus independent of n, and the coordinate limits assemble to $f \in \mathcal{F}(Q_\infty)$ with $D(f^{(k)}, f) \to 0$. Completeness gives the Baire property. □

We use the elementary weight estimate of [1, Eq. (2.2)]: for $u, v \in C(Q_\infty, Q_\infty)$ and any coordinate ℓ,

$$\sup_{x \in Q_\infty} \big|\pi_\ell u(x) - \pi_\ell v(x)\big| \le 2^\ell \, d_\infty(u, v), \qquad d_\infty(u, v) \le \sum_{\ell \ge 1} 2^{-\ell} \sup_x |\pi_\ell u(x) - \pi_\ell v(x)|. \tag{1}$$

Imported dense theorem. The single dynamical input we need from [2] is the exact windowed horseshoe theorem on Q_∞, in the windowed-block form, together with the universal block schedule. We state it in the form used below.

Theorem 2.2 (Dense exact windowed horseshoe; [2, Theorem 3.6 and §12])**.** *Let $W_\bullet = (W_n)_{n \ge 1}$ be a sequence of dense open subsets of Q_∞. For every $f \in \mathcal{F}(Q_\infty)$ and every $\eta_0 > 0$ there is $g \in \mathcal{F}(Q_\infty)$ with $D(f, g) < \eta_0$ admitting an exact $W_\bullet$-windowed Hilbert horseshoe: there are $\alpha \in (0, 1/4)$, support levels K_n, (α, K_n)-boxes J_n, and finite-support boxes $B_{n,1}, \dots, B_{n,M_n} \subseteq \mathrm{int}(J_n) \cap W_n$ with $\mathrm{diam}_\rho(B_{n,i}) < \gamma_n \to 0$ and $g_n(B_{n,i}) = J_{n+1}$ for all n, i, organized on a universal block schedule in which every label from the alphabet $\{C_a, D_s, E_{q,m}\}$ occurs on infinitely many blocks. On such a g, for every $r \ge 0$, the tail $\mathrm{tail}_r g$ carries a compact symbolic semifactor onto a full variable-alphabet shift; in particular every finite admissible itinerary of bricks from time $r+1$ onward is realized by an orbit of $\mathrm{tail}_r g$, with each visited brick lying in its prescribed window.*

The only facts we use from Theorem 2.2 are: (i) along an exact windowed horseshoe, *any* prescribed finite sequence of admissible bricks is followed exactly by some orbit (the finite-itinerary realizability built into the symbolic semifactor); and (ii) the schedule guarantees, on every long block of times, a controllable proportion of "proximity-type" bricks C_a that funnel two orbits into a common small box and "separation-type" bricks D_s that force them apart at the fixed scale t_0. We make (ii) quantitative in Section 4.

Companion inputs. From [1] we use only the framework (the metric D, the weight estimate (1), the Baire property) and, in Section 5, the empty-interior phenomenon as a point of contrast; the proofs below are otherwise self-contained.

3 The finite-horizon windowed DC1 witness

Fix a separation scale $t_0 \in (0, 1)$ once and for all.

Definition 3.1 (Finite-horizon windowed DC1 witness)**.** *Let $W_\bullet = (W_n)$ be a sequence of dense open subsets of Q_∞, let $r \ge 0$, $\varepsilon \in (0, 1)$, $0 < \theta < \eta < 1$ and $N \ge 1$. We say $g \in \mathcal{F}(Q_\infty)$ has the* finite-horizon windowed DC1 witness $\mathcal{W}_{r,\varepsilon,t_0,\eta,\theta,N}(W_\bullet)$ *if there exist $x, y \in Q_\infty$ and disjoint $P, S \subseteq \{0, \dots, N-1\}$ with $\#P \ge (1-\eta)N$ and $\#S \ge \theta N$ such that, with $G_r^j = g_{r+j} \circ \cdots \circ g_{r+1}$,*

*(W1) (*proximity*) $\rho\big(G_r^j(x), G_r^j(y)\big) < \varepsilon$ for all $j \in P$;*

(W2) (separation) $\rho\big(G_r^j(x), G_r^j(y)\big) > t_0$ *for all* $j \in S$*;*

(W3) (windows) $G_r^j(x) \in W_{r+j+1}$ *and* $G_r^j(y) \in W_{r+j+1}$ *for all* j *with* $0 \le j < N$*.*

Write $\mathcal{W}_{r,\varepsilon,t_0,\eta,\theta,N}(W_\bullet) \subseteq \mathcal{F}(Q_\infty)$ *for the set of systems with this witness.*

The point of the next lemma is stability with a modulus that is finite for each N and *does not decay with coordinate depth.* This is the structural difference from the depth obstruction of [1, Remark 3.6], where margins shrink like 2^{-N} in the depth N; here N is a *time* horizon and the witness is open with a radius that depends only on the horizon length and the uniform-continuity moduli of the finitely many maps involved.

Lemma 3.2 (Finite-horizon stability: openness)**.** *Suppose* g *has the witness* $\mathcal{W}_{r,\varepsilon,t_0,\eta,\theta,N}(W_\bullet)$ strictly*, meaning there is* $\kappa > 0$ *such that* (W1) *holds with* $\varepsilon - 2\kappa$ *in place of* ε *on* P*,* (W2) *holds with* $t_0 + 2\kappa$ *on* S*, and each window point lies* ρ*-distance* $> 2\kappa$ *inside its window* W_{r+j+1}*. Then there is* $\delta > 0$ *such that every* $h \in \mathcal{F}(Q_\infty)$ *with* $D(h, g) < \delta$ *also has the witness* $\mathcal{W}_{r,\varepsilon,t_0,\eta,\theta,N}(W_\bullet)$*, with the* same x, y, P, S*. In particular* $\mathcal{W}_{r,\varepsilon,t_0,\eta,\theta,N}(W_\bullet)$ *is open in* $(\mathcal{F}(Q_\infty), D)$*.*

Proof. The maps $g_{r+1}, \dots, g_{r+N-1}$ are uniformly continuous; let ω be a common modulus of continuity (with respect to ρ) for all of them, so ω is nondecreasing, $\omega(0^+) = 0$, and $\rho(g_{r+i}(u), g_{r+i}(v)) \le \omega(\rho(u, v))$ for $1 \le i \le N-1$. Fix x (the same argument applies to y) and write $X_j = G_r^j(x)$ for the g-orbit and $X_j^h = H_r^j(x)$ for the h-orbit, where $H_r^j = h_{r+j} \circ \cdots \circ h_{r+1}$. Set $e_j = \rho(X_j^h, X_j)$. Then $e_0 = 0$ and, for $1 \le j < N$,

$$e_j = \rho\big(h_{r+j}(X_{j-1}^h), g_{r+j}(X_{j-1})\big) \le \rho\big(h_{r+j}(X_{j-1}^h), g_{r+j}(X_{j-1}^h)\big) + \rho\big(g_{r+j}(X_{j-1}^h), g_{r+j}(X_{j-1})\big) \le D(h, g) + \omega(e_{j-1}).$$

Writing $\delta_0 = D(h, g)$, the recursion $e_j \le \delta_0 + \omega(e_{j-1})$, $e_0 = 0$, gives a bound $e_j \le E_j(\delta_0)$ where $E_0 = 0$, $E_j(\delta_0) = \delta_0 + \omega(E_{j-1}(\delta_0))$. Each E_j is a finite composition of the continuous nondecreasing maps $s \mapsto \delta_0 + \omega(s)$ evaluated at the continuous-in-δ_0 starting value, hence $E_j(\delta_0) \to 0$ as $\delta_0 \to 0$ for every fixed j. Choose $\delta > 0$ so small that $E_{N-1}(\delta) < \kappa$; this is possible because there are only finitely many indices $j \le N-1$ and each E_j is continuous at 0 with $E_j(0) = 0$. Then for $D(h, g) < \delta$ we have $\rho(X_j^h, X_j) < \kappa$ and, identically, $\rho(Y_j^h, Y_j) < \kappa$, for all $j < N$.

It remains to transfer (W1)–(W3). For $j \in P$,

$$\rho(X_j^h, Y_j^h) \le \rho(X_j^h, X_j) + \rho(X_j, Y_j) + \rho(Y_j, Y_j^h) < \kappa + (\varepsilon - 2\kappa) + \kappa = \varepsilon,$$

so (W1) holds for h. For $j \in S$,

$$\rho(X_j^h, Y_j^h) \ge \rho(X_j, Y_j) - \rho(X_j, X_j^h) - \rho(Y_j, Y_j^h) > (t_0 + 2\kappa) - \kappa - \kappa = t_0,$$

so (W2) holds. For (W3), each g-orbit point lies ρ-distance $> 2\kappa > \kappa$ inside its open window, and $\rho(X_j^h, X_j) < \kappa$, so $X_j^h \in W_{r+j+1}$, and likewise for y. Thus h has the witness with the same data, proving openness; the strict witness for g is itself an instance, so the open set is nonempty when populated as in Section 4. □

Remark 3.3 (Depth-freeness, made explicit)**.** *The bound* $e_j \le \delta_0 + \omega(e_{j-1})$ *never refers to a coordinate index: the only inputs are the metric distances and the moduli* ω *of the finitely many maps* $g_{r+1}, \dots, g_{r+N-1}$*. Consequently the stability radius* δ *of Lemma 3.2 is governed by the* horizon N *and the maps' moduli, not by how deep a coordinate the proximity scale* ε *or*

the separation scale t_0 resolves. Contrast [1, Remark 3.6]: a covering condition reading image coordinate N has stability $\Theta(2^{-N})$ in the depth N*, because by (1) a D-perturbation of size δ_0 moves that coordinate by up to $2^N\delta_0$. No such factor enters here, because (W1)–(W3) constrain ρ between two orbits rather than a single deep coordinate of one image. The finite-horizon layer is therefore stable precisely where the infinite-depth covering layer is not.*

Definition 3.4 (Finite-horizon scrambled packet)**.** *Let $M \geq 2$, let $W_\bullet = (W_n)$, let $r \geq 0$, $\varepsilon \in (0,1)$, $0 < \eta < 1$, and $N \geq 1$. We say $g \in \mathcal{F}(Q_\infty)$ has the $(M, r, \varepsilon, \eta, N)$-*finite-horizon scrambled packet *if there are points $x^1, \dots, x^M \in Q_\infty$, a set $P \subseteq \{0, \dots, N-1\}$, and sets $S_{uv} \subseteq \{0, \dots, N-1\}$, $1 \leq u < v \leq M$, such that, with $G_r^j = g_{r+j} \circ \cdots \circ g_{r+1}$,*

(P1) $\#P \geq (1-\eta)N$;

(P2) $\#S_{uv} \geq (\eta/8)N$ for every $1 \leq u < v \leq M$;

(P3) $\rho(G_r^j(x^u), G_r^j(x^v)) < \varepsilon$ for every $j \in P$ and every $u < v$;

(P4) $\rho(G_r^j(x^u), G_r^j(x^v)) > t_0$ for every $j \in S_{uv}$;

(P5) $G_r^j(x^u) \in W_{r+j+1}$ for every $u \leq M$ and every $0 \leq j < N$.

Write

$$\mathcal{P}_{M,r,\varepsilon,\eta,N}(W_\bullet) \subseteq \mathcal{F}(Q_\infty)$$

for the set of systems with such a packet.

Lemma 3.5 (Finite-packet stability)**.** *If g has the packet of Definition 3.4 with all inequalities strict by a common margin $\kappa > 0$, then there is $\delta > 0$ such that every $h \in \mathcal{F}(Q_\infty)$ with $D(h, g) < \delta$ has the same packet, with the same points x^u and the same sets P, S_{uv}. In particular $\mathcal{P}_{M,r,\varepsilon,\eta,N}(W_\bullet)$ is open.*

Proof. The proof is the same orbit-tracking argument as Lemma 3.2. Let ω be a common modulus of continuity for the finitely many maps $g_{r+1}, \dots, g_{r+N-1}$. For each packet point x^u, the orbit error under a perturbation h obeys

$$e_j^u \leq D(h, g) + \omega(e_{j-1}^u), \qquad e_0^u = 0.$$

Choose δ so that all errors are $< \kappa$ for $j < N$. Then every proximity inequality loses at most 2κ, every separation inequality loses at most 2κ, and every window visit remains inside its open window. The counting conditions are unchanged. □

Lemma 3.6 (Binary packet codes)**.** *For every $M \geq 2$ there is L_M such that for every $L \geq L_M$ one can choose words $c^1, \dots, c^M \in \{+,-\}^L$ with pairwise Hamming distance at least $L/4$.*

Proof. Choose M independent uniform words of length L. For a fixed pair, the Hamming distance has distribution $\mathrm{Bin}(L, 1/2)$. By the Chernoff bound,

$$\mathbb{P}\{d_H(c^u, c^v) < L/4\} \leq e^{-L/16}.$$

Thus the probability that some pair violates the $L/4$ bound is at most $\binom{M}{2} e^{-L/16}$, which is < 1 once L is large enough. Hence a choice with the desired distances exists. □

4 Density and the residual theorem

We now show each witness set is dense and assemble Theorem 1.1. The density input is Theorem 2.2; the new ingredient is a block schedule that places enough proximity blocks and separation blocks before a common horizon to realize the fraction constraints $\#P \geq (1-\eta)N$, $\#S \geq \theta N$, with $0 < \theta < \eta < 1$.

Block schedule. On the universal schedule of Theorem 2.2 the time axis $\{1, 2, \dots\}$ is partitioned into consecutive blocks $\mathbf{B}_1, \mathbf{B}_2, \dots$, where block $\mathbf{B}_k$ occupies L_k consecutive times and carries a single label. We use two kinds of labelled bricks supplied by the construction of [2, §12]:

- a *proximity brick* C of diameter $< \gamma$, with the property that any two orbits forced through C at a given time are afterwards ρ-distance $< \gamma$ apart at that time (orbits entering a common small box are close);
- a *separation brick* D consisting of two sub-bricks D^+, D^- separated in the first q low coordinates, where $q = q(t_0)$ is any integer with $1 - 2^{-q} > t_0$: the construction of [2, §12] supplies, for each such q, sub-bricks placed in opposite corners of the first q coordinate intervals, so that $|u_i - v_i| = 1$ for every coordinate $i \leq q$ whenever $u \in D^+$, $v \in D^-$. Then $\rho(u, v) \geq \sum_{i \leq q} 2^{-i} |u_i - v_i| = \sum_{i \leq q} 2^{-i} = 1 - 2^{-q} > t_0$, so routing one orbit into D^+ and the other into D^- forces $\rho > t_0$ at that time.

Because every label occurs on infinitely many blocks, for any prescribed long horizon we may select, by the finite-itinerary realizability of Theorem 2.2, an admissible itinerary that spends a chosen number of times in proximity bricks and a chosen number in separation sub-bricks, all inside the prescribed windows. The only arithmetic we need is that the block lengths can be made a vanishing fraction of the horizon, so the fractions $1 - \eta$ and θ are attainable with room for the slack $\eta - \theta$.

Lemma 4.1 (Schedule arithmetic)**.** *Let $0 < \theta < \eta < 1$. There is a sequence of horizons $N \to \infty$ and, for each such N, disjoint sets $P, S \subseteq \{0, \dots, N-1\}$ with $\#P \geq (1-\eta)N$ and $\#S \geq \theta N$ such that P is a union of full proximity blocks and S is a union of full separation blocks of the schedule (each block placed at times $< N$), provided the block lengths satisfy $L_k/(L_1 + \cdots + L_k) \to 0$.*

Proof. Choose the schedule so that proximity and separation blocks alternate in long runs. Write $T_k = L_1 + \cdots + L_k$. Allocate to S the separation blocks and to P the proximity blocks among $\mathbf{B}_1, \dots, \mathbf{B}_k$, with the run lengths chosen so that the total separation length σ_k satisfies $\theta T_k \leq \sigma_k \leq \theta T_k + L_k$. Then

$$\#S \geq \theta T_k, \qquad \#P \geq (1-\theta) T_k - L_k.$$

Setting $N = T_k$, the separation estimate gives $\#S \geq \theta N$. Since $\theta < \eta$ and $L_k/T_k \to 0$, for all sufficiently large k we also have

$$(1-\theta)N - L_k \geq (1-\eta)N.$$

Thus $\#P \geq (1-\eta)N$ along $N = T_k \to \infty$. Disjointness of P, S is automatic since each is a union of distinct whole blocks. □

Lemma 4.2 (Density of the witness set)**.** *For every $W_\bullet$, every $r \geq 0$, every $\varepsilon \in (0,1)$, and every $0 < \theta < \eta < 1$, there are arbitrarily large horizons N for which $\mathcal{W}_{r,\varepsilon,t_0,\eta,\theta,N}(W_\bullet)$ is dense in $(\mathcal{F}(Q_\infty), D)$.*

Proof. Fix $f \in \mathcal{F}(Q_\infty)$ and $\eta_0 > 0$; we find g with $D(f,g) < \eta_0$ in the witness set. Apply Theorem 2.2 to the window sequence $W_\bullet$ to obtain $g \in \mathcal{F}(Q_\infty)$, $D(f,g) < \eta_0$, carrying an exact $W_\bullet$-windowed horseshoe on the universal schedule, with proximity bricks C of diameter $\gamma < \varepsilon$ and separation bricks $D = D^+ \cup D^-$ separated in the first $q = q(t_0)$ coordinates (Remark 4.3). Fix a tail $r \geq 0$. By Lemma 4.1 choose a horizon N and disjoint $P, S \subseteq \{0, \dots, N-1\}$, with $\#P \geq (1-\eta)N$ and $\#S \geq \theta N$, such that P is a union of proximity blocks and S is a union of separation blocks at times $< N$.

Build the two itineraries from time $r+1$. Let both orbits visit the *same* proximity brick at every time $j \in P$, and let one orbit visit D^+ and the other D^- at every time $j \in S$; at the remaining times $j < N$ (within blocks that are neither wholly in P nor in S, of total length $o(N)$) let both orbits follow any common admissible brick, which we also count into the windows but into neither P nor S. This is an admissible finite itinerary of bricks, all lying in their prescribed windows W_{r+j+1}. By the finite-itinerary realizability of Theorem 2.2 there exist $x, y \in Q_\infty$ whose tail_r-orbits follow these itineraries exactly for $0 \leq j < N$; in particular $G_r^j(x), G_r^j(y) \in W_{r+j+1}$ for all $j < N$, giving (W3).

For $j \in P$ both orbit points lie in the same proximity brick of diameter $\gamma < \varepsilon$, so $\rho(G_r^j(x), G_r^j(y)) \leq \gamma < \varepsilon$, giving (W1). For $j \in S$ the points lie in D^+ and D^- respectively. These sub-bricks are built in [2, §12] separated in each of the first q coordinates, so $|\pi_i G_r^j(x) - \pi_i G_r^j(y)| = 1$ for every coordinate index $i \leq q$, and therefore $\rho(G_r^j(x), G_r^j(y)) \geq \sum_{i \leq q} 2^{-i} = 1 - 2^{-q} > t_0$, giving (W2). Thus g has the witness $\mathcal{W}_{r,\varepsilon,t_0,\eta,\theta,N}(W_\bullet)$ and $D(f,g) < \eta_0$. Since f, η_0 were arbitrary, the witness set is dense. □

Remark 4.3 (Calibrating t_0 to the separation bricks)**.** *The separation scale t_0 in Definition 3.1 is fixed in advance. A single first-coordinate gap contributes at most $2^{-1} \cdot 1 = \frac{1}{2}$ to ρ, so separating in coordinate 1 alone caps the achievable separation at $\frac{1}{2}$ and only handles $t_0 < \frac{1}{2}$. To reach any $t_0 \in (0,1)$ we separate in the first q low coordinates: fix an integer $q = q(t_0)$ with $1 - 2^{-q} > t_0$, and use the separation bricks of [2, §12] whose sub-bricks D^+, D^- sit in opposite corners of the first q coordinate intervals (the dist_q construction there supplies exactly these), so that $\rho \geq \sum_{i \leq q} 2^{-i} = 1 - 2^{-q} > t_0$ at separation times. Fully separating the first q coordinates reaches $1 - 2^{-q}$, which exceeds any $t_0 < 1$ once q is large enough. The strict-witness hypothesis of Lemma 3.2 is met because $1 - 2^{-q} > t_0$ by a positive margin and the proximity diameter is $< \varepsilon$ by a positive margin, with window points in the interiors of the (open) windows; one fixes $\kappa > 0$ smaller than all these margins.*

Lemma 4.4 (Density of finite scrambled packets)**.** *For every $M \geq 2$, every $W_\bullet$, every $r \geq 0$, every $\varepsilon \in (0,1)$, and every $0 < \eta < 1$, there are arbitrarily large horizons N for which $\mathcal{P}_{M,r,\varepsilon,\eta,N}(W_\bullet)$ is dense in $(\mathcal{F}(Q_\infty), D)$.*

Proof. Fix $f \in \mathcal{F}(Q_\infty)$ and $\eta_0 > 0$. Apply Theorem 2.2 to get g with $D(f,g) < \eta_0$ carrying an exact windowed horseshoe. Choose the proximity bricks with diameter $\gamma < \varepsilon$, and choose separation bricks D^+, D^- calibrated to t_0 as in Remark 4.3.

Use Lemma 4.1 with the pair (θ, η) replaced by $(\eta/2, \eta)$. Along arbitrarily large horizons N this gives disjoint sets $P, S \subseteq \{0, \dots, N-1\}$ such that P is a union of proximity blocks, S is a

union of separation blocks,

$$\#P \geq (1-\eta)N, \qquad \#S \geq (\eta/2)N.$$

Taking N still larger if necessary, assume $\#S \geq L_M$, where L_M is from Lemma 3.6. Choose words

$$c^1, \ldots, c^M \in \{+,-\}^S$$

with pairwise Hamming distance at least $\#S/4$.

Build M finite itineraries from time $r+1$. At every time $j \in P$, all M itineraries visit the same proximity brick. At every separation time $j \in S$, the u-th itinerary visits D^+ or D^- according to the symbol c_j^u. At all remaining times below N, all itineraries follow any common admissible brick. Finite-itinerary realizability gives points $x^1, \ldots, x^M$ whose tail-r orbits follow these itineraries.

All orbit points lie in the prescribed windows. On P, all points lie in the same proximity brick, so all pairwise distances are $< \varepsilon$. For each pair $u < v$, let

$$S_{uv} := \{j \in S : \ c_j^u \neq c_j^v\}.$$

Then $\#S_{uv} \geq \#S/4 \geq (\eta/8)N$, and at every $j \in S_{uv}$ the two orbits lie in opposite separation sub-bricks, hence are t_0-separated. Thus $g \in \mathcal{P}_{M,r,\varepsilon,\eta,N}(W_\bullet)$, proving density. □

Theorem 4.5 (Residual finite-horizon windowed DC1 witnesses)**.** *Let $\{W_\bullet^a\}_{a\geq 1}$ be a countable family of sequences of dense open subsets of Q_∞ and let $t_0 \in (0,1)$. Then*

$$\mathcal{G} = \bigcap_{a\geq 1} \bigcap_{r\geq 0} \bigcap_{\substack{\varepsilon\in\mathbb{Q}\\ 0<\varepsilon<1}} \bigcap_{\substack{\eta,\theta\in\mathbb{Q}\\ 0<\theta<\eta<1}} \bigcap_{N_0\geq 1} \bigcup_{N\geq N_0} \mathcal{W}_{r,\varepsilon,t_0,\eta,\theta,N}(W_\bullet^a)$$

is comeager in $(\mathcal{F}(Q_\infty), D)$, and every $g \in \mathcal{G}$ satisfies the conclusion of Theorem 1.1.

Proof. For fixed $a, r, \varepsilon, \eta, \theta, N_0$ the set

$$U_{N_0}^{a,r,\varepsilon,\eta,\theta} = \bigcup_{N\geq N_0} \mathcal{W}_{r,\varepsilon,t_0,\eta,\theta,N}(W_\bullet^a)$$

is open by Lemma 3.2 (a union of open sets) and dense: by Lemma 4.2 there are arbitrarily large horizons N for which $\mathcal{W}_{r,\varepsilon,t_0,\eta,\theta,N}(W_\bullet^a)$ is dense, so in particular some $N \geq N_0$ gives a dense subset of $U_{N_0}^{a,r,\varepsilon,\eta,\theta}$, whence the union is dense. The index set of all $a, r, N_0 \in \mathbb{N}$, rational $\varepsilon \in (0,1)$, and rational $0 < \theta < \eta < 1$ is countable, so $\mathcal{G}$ is a countable intersection of open dense sets, hence comeager by Lemma 2.1. If $g \in \mathcal{G}$ then for every a, r, every rational $\varepsilon \in (0,1)$, and all rational $0 < \theta < \eta < 1$, for every N_0 there is $N \geq N_0$ with $g \in \mathcal{W}_{r,\varepsilon,t_0,\eta,\theta,N}(W_\bullet^a)$, i.e. there are x, y, P, S as in Definition 3.1 for that N; this is exactly the statement of Theorem 1.1 (for irrational ε pass to a smaller rational, which only strengthens (W1); the tolerances η, θ are required rational, as stated). □

Theorem 4.6 (Residual finite-horizon scrambled packets)**.** *Let $\{W_\bullet^a\}_{a\geq 1}$ be a countable family of sequences of dense open subsets of Q_∞ and let $t_0 \in (0,1)$. Then*

$$\mathcal{G}_{\mathrm{pkt}} = \bigcap_{a\geq 1} \bigcap_{r\geq 0} \bigcap_{M\geq 2} \bigcap_{\substack{\varepsilon\in\mathbb{Q}\\ 0<\varepsilon<1}} \bigcap_{\substack{\eta\in\mathbb{Q}\\ 0<\eta<1}} \bigcap_{N_0\geq 1} \bigcup_{N\geq N_0} \mathcal{P}_{M,r,\varepsilon,\eta,N}(W_\bullet^a)$$

is comeager in $(\mathcal{F}(Q_\infty), D)$. Every $g \in \mathcal{G}_{\mathrm{pkt}}$ satisfies the packet conclusion stated in Theorem 1.2.

Proof. For fixed $a, r, M, \varepsilon, \eta, N_0$, the union over $N \geq N_0$ is open by Lemma 3.5 and dense by Lemma 4.4. The displayed intersection is countable, hence comeager. Membership gives arbitrarily large horizons with the packet of Definition 3.4. Passing from rational ε, η to the stated rational parameters is immediate. □

Remark 4.7 (What the witnesses approximate)**.** *Fix $g \in \mathcal{G}$, a tail r and a scale ε. Taking $\eta = 1/m$ and $\theta = 1/(2m)$ for $m = 2, 3, \ldots$ yields, for each m, arbitrarily long horizons carrying a pair proximal on a $(1 - 1/m)$-fraction of times and t_0-separated on at least a $1/(2m)$-fraction. This is the finite-horizon, fixed-tolerance silhouette of a DC1 scrambled pair: the upper proximity statistic tends to 1 and the separation statistic stays positive at the fixed scale t_0, with all visits inside the prescribed windows up to the horizon. We do not assert a single pair valid for all horizons at once; producing such a pair is the infinite-horizon problem (Problem 6.1), which Section 5 shows no finite-horizon certificate can reach.*

Remark 4.8 (Finite packets are still finite-horizon)**.** *Theorem 4.6 strengthens the positive side from one pair to arbitrarily large finite packets. It is still a finite-horizon statement: the packet points and the code across the separation times may depend on N, and the theorem does not produce one uncountable scrambled set, or even one finite packet, working along all horizons simultaneously. The binary coding only says that the dense horseshoe's two separation sub-bricks can be reused over many separation times to make every pair in a finite list disagree often.*

5 The obstruction: no finite-horizon certificate suffices

We now prove Theorem 1.3. We first recall metric mean dimension, since the squeeze argument controls it directly and infinite topological entropy is one of its consequences in the relevant direction.

For a non-autonomous system h on Q_∞, a tail $\mathrm{tail}_s h$, scale $\varepsilon > 0$ and horizon N, let $\mathrm{sep}_{s,N}(h, \varepsilon)$ be the maximal cardinality of a set $E \subseteq Q_\infty$ that is (s, N, ε)-separated in the Bowen metric $\rho^h_{s,N}(x, y) = \max_{0 \leq j < N} \rho(H^j_s(x), H^j_s(y))$, and put

$$\mathrm{mdim}_{\mathrm{M}}(\mathrm{tail}_s h) = \limsup_{\varepsilon \downarrow 0} \frac{1}{\log(1/\varepsilon)} \limsup_{N \to \infty} \frac{1}{N} \log \mathrm{sep}_{s,N}(h, \varepsilon).$$

The inner double limit is the topological entropy $h_{\mathrm{top}}(\mathrm{tail}_s h)$ at scale ε; if $h_{\mathrm{top}}(\mathrm{tail}_s h) = \infty$ then $\frac{1}{N} \log \mathrm{sep}_{s,N} \to \infty$ as $\varepsilon \downarrow 0$ along the inner limit, and one checks $\mathrm{mdim}_{\mathrm{M}}$ may still be finite or infinite. We only use the following one-directional bound.

Lemma 5.1 (Squeeze cap)**.** *Let $L \geq 1$ and let $h \in \mathcal{F}(Q_\infty)$ satisfy $\pi_j h_n(x) = \frac{1}{2}$ for every n, every $x \in Q_\infty$, and every coordinate $j > L$ — that is, every image of every h_n lies in the L-dimensional slice $Z_L = \{z \in Q_\infty : \ z_j = \frac{1}{2} \ \forall j > L\}$. Then for every tail $s \geq 0$ and every fixed scale $0 < \varepsilon \leq 1$,*

$$h^{(\varepsilon)}_{\mathrm{top}}(\mathrm{tail}_s h) := \limsup_{N \to \infty} \frac{1}{N} \log \mathrm{sep}_{s,N}(h, \varepsilon) \ \leq \ L \log(4/\varepsilon), \tag{2}$$

so the entropy at every fixed scale is finite and $\mathrm{mdim}_{\mathrm{M}}(\mathrm{tail}_s h) \leq L < \infty$.

Proof. Fix $s \geq 0$. Since every h_n maps into Z_L, for every $i \geq 1$ and every x the point $H_s^i(x)$ lies in Z_L; only the base point $H_s^0(x) = x$ can lie outside Z_L. The slice Z_L is isometric, in the metric ρ restricted to it, to $[0,1]^L$ with the weighted metric $d_L(u,v) = \sum_{j \leq L} 2^{-j}|u_j - v_j|$, because coordinates $j > L$ are constant on Z_L. A grid of mesh $\varepsilon/2$ in each of the first L coordinates has $(\lceil 2/\varepsilon \rceil)^L$ cells and resolves d_L to within $\sum_{j \leq L} 2^{-j}(\varepsilon/2) < \varepsilon/2 < \varepsilon$; hence the ρ-covering number of Z_L at radius ε satisfies

$$\operatorname{cov}_\rho(Z_L, \varepsilon) \;\leq\; (\lceil 2/\varepsilon \rceil)^L \;\leq\; (4/\varepsilon)^L \qquad (\varepsilon \leq 1).$$

Now bound the Bowen separation. An (s, N, ε)-separated set E is, in particular, mapped injectively by $x \mapsto (H_s^0(x), H_s^1(x), \ldots, H_s^{N-1}(x))$ to a set that is ε-separated in the max-ρ metric on the product $Q_\infty \times Z_L \times \cdots \times Z_L$ ($N-1$ copies of Z_L). An ε-separated set in a product (max metric) has cardinality at most the product of the factors' ε-covering numbers, so

$$\operatorname{sep}_{s,N}(h, \varepsilon) \;\leq\; \operatorname{cov}_\rho(Q_\infty, \varepsilon) \cdot \operatorname{cov}_\rho(Z_L, \varepsilon)^{N-1} \;\leq\; \operatorname{cov}_\rho(Q_\infty, \varepsilon) \cdot (4/\varepsilon)^{L(N-1)}.$$

The factor $\operatorname{cov}_\rho(Q_\infty, \varepsilon)$ is a finite constant depending only on ε (the Hilbert cube is compact), and crucially it is independent of N. Therefore

$$\frac{1}{N} \log \operatorname{sep}_{s,N}(h, \varepsilon) \;\leq\; \frac{\log \operatorname{cov}_\rho(Q_\infty, \varepsilon)}{N} + \frac{N-1}{N} L \log(4/\varepsilon) \;\xrightarrow[N \to \infty]{}\; L \log(4/\varepsilon).$$

Thus (2) holds, the right-hand side is finite for each fixed $0 < \varepsilon \leq 1$, and dividing by $\log(1/\varepsilon)$ and letting $\varepsilon \downarrow 0$,

$$\operatorname{mdim}_{\mathrm{M}}(\operatorname{tail}_s h) = \limsup_{\varepsilon \downarrow 0} \frac{h_{\mathrm{top}}^{(\varepsilon)}(\operatorname{tail}_s h)}{\log(1/\varepsilon)} \;\leq\; \limsup_{\varepsilon \downarrow 0} \frac{L \log(4/\varepsilon)}{\log(1/\varepsilon)} = L. \qquad \square$$

Remark 5.2 (Why deep coordinates are collapsed, not merely confined)**.** *It is essential that the squeeze sends deep coordinates to a single value rather than into a proper subinterval. A perturbation confining coordinates $j > L$ to a band $[\frac{1}{8}, \frac{7}{8}]$ would leave those coordinates free to vary, and the band $\prod_{j>L}[\frac{1}{8}, \frac{7}{8}]$ is still an infinite-dimensional Hilbert-cube tail: a map into it can have arbitrarily large metric mean dimension, so confinement to a band does* not *cap* $\operatorname{mdim}_{\mathrm{M}}$*. Collapsing coordinates $j > L$ to the constant $\frac{1}{2}$ lands every image in the genuinely L-dimensional slice Z_L, of finite box dimension L, which is what forces (2). The D-cost is the same to leading order: $\sum_{j>L} 2^{-j}|y_j - \frac{1}{2}| \leq \frac{1}{2}2^{-L}$.*

Remark 5.3 (Which complexity notion the cap contradicts)**.** *We phrase the obstruction through metric mean dimension, which is unambiguous across non-autonomous entropy conventions: (2) gives $\operatorname{mdim}_{\mathrm{M}}(\operatorname{tail}_s h) \leq L < \infty$ for every s. The "infinite topological entropy" tails produced by [2] satisfy the stronger scale law $h_{\mathrm{top}}^{(\varepsilon)}(\operatorname{tail}_s g) \geq q \log(1/\varepsilon) - C_q$ for all q, equivalently $\operatorname{mdim}_{\mathrm{M}} = \infty$; that is the property the squeeze defeats, since (2) caps the same scale-rate by $L \log(4/\varepsilon)$, which cannot exceed $q \log(1/\varepsilon) - C_q$ for all small ε once $q > L$. Thus Theorem 5.4 is proved for the metric-mean-dimension form, and the entropy form follows from the same inequality.*

Theorem 5.4 (Finite-horizon certificate obstruction)**.** *Let $V \subseteq \mathcal{F}(Q_\infty)$ be finite-horizon certified: there are $r \geq 0$ and $N \geq 1$ such that whether $g \in V$ depends only on $(g_{r+1}, \ldots, g_{r+N})$. Suppose*

$$V \subseteq \big\{\, g \in \mathcal{F}(Q_\infty) : \;\operatorname{mdim}_{\mathrm{M}}(\operatorname{tail}_s g) = \infty \text{ for some } s \geq 0 \,\big\}.$$

Then $V = \varnothing$. The same conclusion holds if V is contained in the class of systems some tail of which has infinite topological entropy in the scale-law sense of [2] (i.e. $h_{\mathrm{top}}^{(\varepsilon)} \geq q \log(1/\varepsilon) - C_q$ for all q).

Proof. Suppose $V \neq \varnothing$ and fix $g \in V$. Since V is open and finite-horizon certified by the window $(g_{r+1}, \dots, g_{r+N})$, there is $\delta > 0$ such that every $h \in \mathcal{F}(Q_\infty)$ with $h_{r+i} = g_{r+i}$ for $1 \le i \le N$ and $D(h, g) < \delta$ lies in V; indeed, membership ignores all coordinates outside the window, so any h agreeing with g on the window is in V regardless of its other coordinates, provided it is within D-distance δ of g where δ is the openness radius (which only constrains the windowed coordinates, on which h equals g, so any such h qualifies).

Choose $L \ge 1$ with $2^{-L} < \delta$. Define the coordinate squeeze $\Phi : Q_\infty \to Q_\infty$, collapsing every coordinate past depth L to the constant $\frac{1}{2}$:

$$\Phi(y)_j = \begin{cases} y_j, & j \le L, \\ \frac{1}{2}, & j > L. \end{cases}$$

Φ is continuous, $\Phi(Q_\infty) = Z_L := \{z \in Q_\infty : \ z_j = \frac{1}{2} \ \forall j > L\}$, and

$$\rho(\Phi(y), y) = \sum_{j>L} 2^{-j} \left| y_j - \tfrac{1}{2} \right| \le \sum_{j>L} 2^{-j} \cdot \tfrac{1}{2} = \tfrac{1}{2}\, 2^{-L} < \delta.$$

Define $h \in \mathcal{F}(Q_\infty)$ by leaving the certifying window untouched and squeezing every other map:

$$h_n = \begin{cases} g_n, & n \in \{r+1, \dots, r+N\}, \\ \Phi \circ g_n, & \text{otherwise.} \end{cases}$$

Then $d_\infty(h_n, g_n) = 0$ on the window and $d_\infty(h_n, g_n) = \sup_x \rho(\Phi(g_n(x)), g_n(x)) < \delta$ off the window, so $D(h, g) < \delta$, and h agrees with g on the certifying window; hence $h \in V$.

Now examine an arbitrary tail of h. Fix $s \ge 0$ and consider the orbit $H_s^i = h_{s+i} \circ \cdots \circ h_{s+1}$. For $i \ge 1$ the outermost map is h_{s+i}, which is either a squeezed map $\Phi \circ g_{s+i}$ — in which case $H_s^i(x) \in Z_L$ for every x — or, if $s+i \in \{r+1, \dots, r+N\}$, the untouched map g_{s+i}. The window is a set of at most N times, so the second case occurs for at most N values of $i \ge 1$; call this exceptional set $\mathcal{E} \subseteq \{1, \dots\}$, $\#\mathcal{E} \le N$. For all $i \ge 1$ with $i \notin \mathcal{E}$ we have $H_s^i(x) \in Z_L$ for every x.

Bound the Bowen separation exactly as in Lemma 5.1, isolating the finitely many exceptional time-slices. An (s, N', ε)-separated set injects, via $x \mapsto (H_s^i(x))_{0 \le i < N'}$, into a set that is ε-separated in the max-ρ metric on $\prod_{0 \le i < N'} Y_i$, where $Y_i = Z_L$ for $i \ge 1$, $i \notin \mathcal{E}$, and $Y_i = Q_\infty$ for $i = 0$ or $i \in \mathcal{E}$. Hence

$$\operatorname{sep}_{s,N'}(h, \varepsilon) \ \le \ \operatorname{cov}_\rho(Q_\infty, \varepsilon)^{\,1+\#\mathcal{E}} \cdot \operatorname{cov}_\rho(Z_L, \varepsilon)^{\,N'-1-\#\mathcal{E}} \ \le \ \operatorname{cov}_\rho(Q_\infty, \varepsilon)^{\,1+N} \cdot (4/\varepsilon)^{LN'}.$$

The first factor is independent of N' (the window length N is fixed), so dividing by N' and letting $N' \to \infty$ gives, as in (2),

$$h_{\mathrm{top}}^{(\varepsilon)}(\mathrm{tail}_s h) \ \le \ L \log(4/\varepsilon), \qquad \mathrm{mdim}_{\mathrm{M}}(\mathrm{tail}_s h) \ \le \ L < \infty,$$

for every $s \ge 0$. This contradicts $h \in V \subseteq \{\, \mathrm{mdim}_{\mathrm{M}}(\mathrm{tail}_s g) = \infty \text{ for some } s \,\}$. Hence $V = \varnothing$.

For the scale-law entropy form, an $h \in V$ would need some tail with $h_{\mathrm{top}}^{(\varepsilon)} \ge q \log(1/\varepsilon) - C_q$ for all q; but the bound $h_{\mathrm{top}}^{(\varepsilon)}(\mathrm{tail}_s h) \le L \log(4/\varepsilon)$ fails this for $q > L$ at all small ε. So $V = \varnothing$ in that case as well. □

Remark 5.5 (A horizon obstruction, complementary to the depth obstruction)**.** *The squeeze in Theorem 5.4 acts on all maps* outside a fixed window of times*, whereas the empty-interior squeeze*

of [1, Theorem 6.2] acts on all coordinates past a fixed depth *of a single map. The first defeats finite*-horizon *certificates of infinite-horizon complexity; the second defeats open sets requiring infinite*-depth *covering. Both rest on the same elementary fact — a coordinate (or time) tail of total D-weight $< \delta$ can be overwritten without leaving a δ-ball — but they constrain orthogonal method classes. Read together: no open set can certify infinite depth (depth obstruction), and no finite-horizon-certified open set can certify an infinite-*$\mathrm{mdim}_{\mathrm{M}}$ *tail or a scale-law entropy tail (horizon obstruction).*

Corollary 5.6 (Banach–Mazur form)**.** *Let V be finite-horizon certified and contained in one of the complexity classes covered by Theorem 5.4. Then in the Banach–Mazur game on $(\mathcal{F}(Q_\infty), D)$, Player II (the player choosing the even moves) can avoid V entirely: after any move of Player I, Player II plays a ball produced by the squeeze of Theorem 5.4, all of whose points lie outside V. Equivalently, V is not residual; being also open, V is empty. Consequently no countable union of finite-horizon-certified open sets, each entailing an infinite-*$\mathrm{mdim}_{\mathrm{M}}$ *tail or one of the scale-law entropy tails of Theorem 5.4, can be dense, and the Baire strategy through such a family cannot prove Problem 6.1.*

Proof. By Theorem 5.4, $V = \varnothing$, so every D-ball is disjoint from V; in particular Player II has the trivial avoidance strategy, and the open empty set is vacuously non-residual. The final clause follows because a countable union of empty sets is empty, hence not dense, so its members cannot serve as the dense-open building blocks of a Baire argument. □

The squeeze also defeats semifactors and confines scrambling

The cap of Lemma 5.1 is an entropy statement, and Theorem 5.4 phrases the obstruction through $\mathrm{mdim}_{\mathrm{M}}$ and the scale-law entropy. The same squeezed system h fails a further growth-type conclusion of the dense theorem outright, with no horizon hypothesis: the all-tail symbolic semifactor onto an unbounded-alphabet shift. This sharpens what the method defeats and, by contrast, isolates the one conclusion it leaves standing.

We use the following operational notion, which is the only feature of a symbolic semifactor onto a shift we need. Say a tail $\mathrm{tail}_s h$ admits a *compact symbolic semifactor with branch separation $\varepsilon_* > 0$ and block alphabets $(M_n)_{n\geq 1}$* if for every n the system carries an (s, n, ε_*)-separated set of cardinality at least $\prod_{i\leq n} M_i$. This holds whenever $\mathrm{tail}_s h$ factors onto a one-sided shift whose length-n admissible words number $\prod_{i\leq n} M_i$ and whose distinct length-n cylinders pull back to orbit segments lying in branch boxes pairwise more than ε_* apart in the depth-n Bowen metric $\rho^h_{s,n}$: the factor map sends each length-n Bowen ball into a single cylinder, so distinct admissible words are realized by ε_*-separated segments. The full variable-alphabet shift of Theorem 2.2, with $M_n \to \infty$, is the case of interest.

Lemma 5.7 (Semifactor obstruction)**.** *Let $L \geq 1$ and let $h \in \mathcal{F}(Q_\infty)$ have every image of every h_n in the slice $Z_L = \{z \in Q_\infty : \ z_j = \frac{1}{2} \ \forall j > L\}$. Then no tail* $\mathrm{tail}_s h$ *admits a compact symbolic semifactor with any fixed branch separation $\varepsilon_* > 0$ and block alphabets $M_n \to \infty$; in particular no tail factors onto the full variable-alphabet shift of Theorem 2.2.*

Proof. Suppose some tail $\mathrm{tail}_s h$ did, with separation ε_* and alphabets $M_n \to \infty$. By definition the system carries, for each n, an (s, n, ε_*)-separated set of cardinality at least $\prod_{i\leq n} M_i$, so

$$\mathrm{sep}_{s,n}(h, \varepsilon_*) \ \geq \ \prod_{i\leq n} M_i, \qquad \frac{1}{n} \log \mathrm{sep}_{s,n}(h, \varepsilon_*) \ \geq \ \frac{1}{n} \sum_{i\leq n} \log M_i.$$

The right-hand side is the Cesàro mean of the sequence $(\log M_i)_i$, which tends to ∞ because $M_i \to \infty$; hence $h_{\mathrm{top}}^{(\varepsilon_*)}(\mathrm{tail}_s h) = \limsup_{N\to\infty} \frac{1}{N} \log \mathrm{sep}_{s,N}(h, \varepsilon_*) = \infty$. But Lemma 5.1 applies to h verbatim and gives $h_{\mathrm{top}}^{(\varepsilon_*)}(\mathrm{tail}_s h) \le L \log(4/\varepsilon_*) < \infty$ at the fixed scale ε_*, a contradiction. Hence no such semifactor exists. The variable-alphabet shift of Theorem 2.2 has $M_n \to \infty$ and fixed branch separation, so it is excluded. □

In the obstruction proof of Theorem 5.4 the squeezed map agrees with g on the certifying window and squeezes elsewhere, so only finitely many tail time-slices escape Z_L. The finite-exception bookkeeping that turned (2) into the in-proof cap applies here too: adjoining the fixed finite set $\mathcal{E}$ of unconstrained Bowen coordinates multiplies $\mathrm{sep}_{s,n}$ by a constant $\le \mathrm{cov}_\rho(Q_\infty, \varepsilon_*)^{1+\#\mathcal{E}}$, which changes $\frac{1}{n} \log \mathrm{sep}_{s,n}$ by $o(1)$. So Lemma 5.7 holds for that h as well, and the squeezed system carries no unbounded-alphabet semifactor on any tail.

Proposition 5.8 (Finite-dimensional confinement of scrambling)**.** *Let $L \ge 1$ and let $h \in \mathcal{F}(Q_\infty)$ have every image of every h_n in Z_L. Fix a tail $s \ge 0$. For every $x \in Q_\infty$ the orbit $H_s^i(x)$ lies in Z_L for all $i \ge 1$; consequently every DC1-scrambled set of $\mathrm{tail}_s h$ has its entire post-initial orbit data inside the L-dimensional slice Z_L, and the proximal/distal distributions that define scrambling are computed from orbits in Z_L.*

Proof. For $i \ge 1$ the outermost map of $H_s^i = h_{s+i} \circ \cdots \circ h_{s+1}$ is h_{s+i}, whose image lies in Z_L; hence $H_s^i(x) \in Z_L$ for every x and every $i \ge 1$. Only the base point $H_s^0(x) = x$ can lie outside Z_L. The distributional functions F_{xy}, F_{xy}^* that define a DC1-scrambled pair are limits of empirical distributions of the distances $\rho(H_s^j(x), H_s^j(y))$ over $0 \le j < N$ as $N \to \infty$; the single term $j = 0$ contributes weight $1/N \to 0$, so these distributions are determined by the Z_L-valued part of the orbits. Thus the scrambling lives in Z_L. □

Remark 5.9 (The squeeze does *not* kill a bare DC1 set)**.** *Proposition 5.8 is a confinement statement, not a destruction statement, and it is worth being exact about the difference. Collapsing coordinates $j > L$ leaves the first L coordinates — in particular coordinate 1 — untouched, and the t_0-separation that drives the horseshoe is read in the low coordinates $i \le q \le L$ (Remark 4.3); the squeeze never touches them. Uncountable DC1 scrambling and continuum cardinality are insensitive to ambient dimension: the one-dimensional slice $Z_1 \cong [0,1]$ already carries perfect (hence continuum-sized) DC1-scrambled sets for suitable interval maps, and confining a scrambled set to Z_L neither reduces its cardinality nor disturbs its proximal/distal statistics. So the coordinate squeeze, which caps entropy and kills the unbounded-alphabet semifactor, does* not *by itself defeat the existence of an uncountable DC1-scrambled set: that conclusion is dimension-stable and survives the collapse. What is proved is only that any such set is finite-dimensionally confined after one step.*

Corollary 5.10 (The conjunction fails on the growth conjuncts only)**.** *Let $V \subseteq \mathcal{F}(Q_\infty)$ be finite-horizon certified and contained in the class of systems some tail of which simultaneously (a) carries an uncountable windowed DC1-scrambled set, (b) has infinite topological entropy in the scale-law sense, and (c) admits a compact symbolic semifactor onto the full variable-alphabet shift. Then $V = \varnothing$. The emptiness is forced by conjuncts* (b) *and* (c) *alone: the squeeze of Theorem 5.4 produces, in every D-ball meeting the certifying window, a system whose tails violate both (b) (by Lemma 5.1) and (c) (by Lemma 5.7), while leaving conjunct (a) attainable in the confined slice Z_L (Proposition 5.8, Remark 5.9).*

Proof. If $V \neq \varnothing$, pick $g \in V$ and form the squeezed h of Theorem 5.4: h agrees with g on the certifying window, $D(h, g) < \delta$, so $h \in V$, yet every tail of h has $h_{\mathrm{top}}^{(\varepsilon)} \leq L \log(4/\varepsilon)$ (Lemma 5.1 with the finite-exception bookkeeping of the theorem) and admits no $M_n \to \infty$ semifactor (Lemma 5.7, likewise). Either failure contradicts $h \in V$, since membership requires (b) and (c) of some tail. Hence $V = \varnothing$. That the obstruction does not pass through (a) is Remark 5.9: the same h can still carry an uncountable scrambled set, now confined to Z_L. □

6 The ledger and what remains open

We restate the open problem of [1] (which itself isolates [2, §14]) and record precisely what the present results add.

Problem 6.1 (Residuality of the infinite-horizon conclusions; [1, Problem 8.1])**.** *Let $W_\bullet$ be a sequence of dense open subsets of Q_∞ and let $\mathcal{C}(W_\bullet)$ be the class of $g \in \mathcal{F}(Q_\infty)$ such that for every $r \geq 0$ the tail $\mathrm{tail}_r g$ admits a compact symbolic semifactor onto a full variable-alphabet shift, a continuum-sized DC1-scrambled set visiting W_{r+j+1} at tail time j, infinite topological entropy, and infinite metric mean dimension. By [2, Theorem 3.6], $\mathcal{C}(W_\bullet)$ is dense. Is $\mathcal{C}(W_\bullet)$ residual?*

Ledger.

(i) *Positive, this paper.* The finite-horizon windowed DC1 witnesses are residual: Theorem 4.5 produces a comeager $\mathcal{G}$ on which, at every tail, every rational scale, every rational $0 < \theta < \eta < 1$ and the fixed separation scale t_0, arbitrarily long witness horizons exist. This is the finite-horizon shadow of the DC1 half of $\mathcal{C}(W_\bullet)$, with $\eta \downarrow 0$ recovering the proximity statistic and t_0 the separation statistic. Theorem 4.6 strengthens this from one pair to arbitrary finite packets: for every M, every tail and every proximal scale there are arbitrarily long horizons with M windowed orbit segments which are mutually proximal on a common $1 - \eta$ fraction and whose every pair is t_0-separated on at least an $\eta/8$ fraction.

(ii) *Negative, growth-type conclusions.* No finite-horizon certificate forces the *growth-type* infinite-horizon conclusions. Theorem 5.4 shows every finite-horizon-certified open set contained in the infinite-$\mathrm{mdim}_{\mathrm{M}}$ class, or in the scale-law entropy class from [2], is empty, and the same squeeze also kills the unbounded-alphabet semifactor (Lemma 5.7); the bundled Corollary 5.10 records that the conjunction of entropy, semifactor and DC1 fails on its entropy and semifactor conjuncts alone, while Corollary 5.6 renders the natural Baire strategy through such certificates impossible.

(ii′) *What the squeeze does* not *defeat.* The method does not defeat the bare existence of an uncountable DC1-scrambled set. By Proposition 5.8 the squeeze only confines scrambling to a finite-dimensional slice Z_L, and DC1 scrambling with continuum cardinality survives such confinement (Remark 5.9). So the coordinate-collapse certificate obstruction separates cleanly: it defeats the entropy/mean-dimension/semifactor (growth-rate) conclusions, but leaves the DC1 cardinality conclusion open to finite-horizon methods, and a different obstruction — or a positive construction — would be needed for that conjunct.

(iii) *Still open.* Whether $\mathcal{C}(W_\bullet)$ is residual remains open. Theorem 5.4 does *not* prove $\mathcal{C}(W_\bullet)$ non-comeager: a comeager set need not be a countable intersection of finite-horizon certified open sets, and an infinite-horizon property may be comeager while admitting no

finite-horizon certificate at all (it would be witnessed by an unbounded, system-dependent window). We make no claim either way.

Where a positive proof must live. Combining (ii) with the depth obstruction of [1, Theorem 6.2]: a residual proof of Problem 6.1 cannot use open building blocks that certify either infinite depth or an infinite-$\mathrm{mdim}_{\mathrm{M}}$/scale-law entropy tail from a bounded window of times. The finite-horizon witnesses of Theorem 4.5 are exactly the strongest windowed objects that *are* residual, and Theorem 4.6 shows that this includes arbitrary finite packet cardinalities. Upgrading these finite packets to a single infinite-horizon scrambled set requires diagonalizing across horizons in a system-dependent way, reading unboundedly many maps to pin one system — precisely the "replacement for the backward induction consuming only finite-depth information per step" requested in [1], now seen to also need to consume unboundedly many *times*. We formulate the sharpened target.

Problem 6.2 (Finite-horizon fusion)**.** *Does there exist a comeager set on which a* single *pair* (x, y) *(depending on the system) is DC1-scrambled at every tail with prescribed window visits at all times — equivalently, on which the finite-horizon witnesses of Theorem 4.5 can be chosen coherently in* N *so as to glue into an infinite-horizon scrambled pair? By Theorem 5.4 any such gluing cannot be certified by a bounded window of times; it must be a genuinely non-finite-horizon (system-dependent, unbounded-window) construction.*

A positive answer to Problem 6.2 for the DC1 component, together with the all-tail semi-factor of Theorem 2.2, would settle the DC1 half of Problem 6.1; the entropy and metric-mean-dimension halves would still require their own non-finite-horizon arguments, by Theorem 5.4. We intend to return to this elsewhere.

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Declaration on generative AI

During the preparation of this work the author used generative AI tools to accelerate drafting and revision. All mathematical claims, proofs, and bibliographic information were subsequently reviewed and edited by the author, who takes full responsibility for the content of the manuscript.

References

[1] Y. Shmalo, *Margin-stable schemes and the Baire category of full complexity in non-autonomous dynamics on the Hilbert cube*, companion manuscript, 2026.

[2] Y. Shmalo, *Strict cubical horseshoes and all-tail full complexity in non-autonomous dynamics*, manuscript, submitted, 2026.

[3] C. Miranda, *Un'osservazione su un teorema di Brouwer*, Boll. Un. Mat. Ital. (2) **3** (1940–41), 5–7.

[4] W. Kulpa, *The Poincaré–Miranda theorem*, Amer. Math. Monthly **104** (1997), no. 6, 545–550.

[5] S. Kolyada and L. Snoha, *Topological entropy of nonautonomous dynamical systems*, Random Comput. Dynam. **4** (1996), 205–233.

[6] F. Balibrea, *On problems of topological dynamics in non-autonomous discrete systems*, Appl. Math. Nonlinear Sci. **1** (2016), no. 2, 391–404.

[7] J. C. Oxtoby, *The Banach–Mazur game and Banach category theorem*, in: Contributions to the Theory of Games, Vol. III, Ann. of Math. Stud. **39**, Princeton Univ. Press, 1957, 159–163.

[8] E. Lindenstrauss and B. Weiss, *Mean topological dimension*, Israel J. Math. **115** (2000), 1–24.

[9] B. Schweizer and J. Smítal, *Measures of chaos and a spectral decomposition of dynamical systems on the interval*, Trans. Amer. Math. Soc. **344** (1994), no. 2, 737–754.